Lab - Research a Hardware Upgrade

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CPE11S4

Introduction

Use the Internet, a newspaper, or a local store to gather information about hardware components. Your customer's computer currently has one module of 2 GB of RAM, a 500 GB hard disk drive, and a PCIe video adapter card with 256 MB of RAM. Your customer wants to be able to play advanced video games.

Instructions

Step 1: Research memory options.

Shop around, and in the table below list the brand, model number, features, and cost for two different 8 GB modules of DDR3.

Brand and Model Number	Features	Cost
Kingston ValueRAM (KVR16N11H/8)	*It is running at 1600MHz (PC3-12800) with a default CAS latency of CL11 and operating on the JEDEC-standard 1.5V power supply.*	*₱ 2,860.00*
Corsair Vengeance (CMZ8GX3M1A1600C10)	*It operates at 1600MHz (PC3-12800) with CL10 (10-10-10-27) timings and a 1.5V operating voltage.*	*₱ 1,250.00*

Question:

Based on your research, which RAM would you select? Be prepared to discuss your decisions regarding the RAM you select.

- Based on the features that I have researched, yes, they are both 8GB DDR3 memory modules running at 1600MHz with a 1.5V operating voltage. However, the Corsair Vengeance offers a lower CAS latency of CL10, compared to the Kingston Value RAM's default CL11, resulting in slightly faster memory response times. It also includes a dedicated aluminum heat spreader, which helps dissipate heat during extended use and makes it more suitable for gaming, multimedia editing, and other demanding workloads.

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Step 2: Research hard disk drive options.

Shop around, and in the table below list the brand, model number, features, and cost for a 3 TB 7200 rpm SATA 3 hard disk drive and a 500GB or 1TB solid state drive (SSD).

Brand and Model Number	Features	Cost
3 TB HDD Seagate BarraCuda (ST3000DM008)	*It features a 7,200 RPM spindle speed and a 256MB cache, and connects through a SATA 6Gb/s interface.*	*₱ 9,779.00*
1 TB SSD Samsung 980 (MZ-V8V1T0)	*One of its features is the speed where it reads up to 3,500 MB/s and utilizes the PCIe 3.0 x4 interface and NVMe technology.*	*₱ 5,299.00*

Question:

Based on your research, which hard disk drive would you select? Be prepared to discuss your decisions regarding the hard disk drive you select.

- Choosing between the two hard disk drives, I would choose the Samsung 980 1TB SSD (MZ-V8V1T0) because it offers significantly faster performance than the Seagate BarraCuda 3TB HDD (ST3000DM008). The SSD provides much quicker boot times, faster application loading, and improved file transfer speeds because of its PCIe 3.0 x4 NVMe interface.

Step 3: Research video adapter card options.

Shop around, and in the table below list the brand, model number, features, and cost for two different PCIe video adapter cards with at least 1 GB of RAM.

Brand and Model Number	Features	Cost
BlankXFX Radeon HD 6850 (HD-685X-ZCFC)	*It gives a 960 stream processors, a 775 MHz core clock, and 1GB of GDDR5 memory on a 256-bit memory interface.*	*₱ 2,459.00*
BlankPowerColor Radeon RX 570 (AXRX 570 4GBD5-3DHD/OC)	*This features a 2,048 stream processors, a boost clock of up to 1,250 MHz, and 4GB of GDDR5 memory with a 256-bit memory interface.*	*₱ 5,949.00*

Question:

Based on your research, which video adapter card would you select? Be prepared to discuss your decisions regarding the video adapter card you select.

- Base on my research, both graphics cards are capable of handling everyday computing and multimedia tasks, the BlankPowerColor Radeon RX 570 offers significantly better overall performance. Another reason why I chose it because of its improved cooling system creating a video adapter card that does not heat a lot.

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